ARCHITECTURE BRIEF

# Company-Wide AI Assistant

Extending the NPS dashboard assistant to answer questions across documents and structured company data

Hatch Africa | Prepared by Daniel Temesgen | 2 October 2026 | Internal use only

## Executive summary

Hatch already runs a production AI assistant inside the Agent NPS dashboard. It is small, but it is built on the right principle, and that principle is what should govern a company-wide rollout.

> **The language model selects and composes. It never calculates, and it is never the access-control boundary.** Numbers come from the database; permissions come from the user's identity. Everything in this brief is machinery for holding that line at company scale.

- **What exists today.** A constrained tool-calling assistant over a single NPS dataset of roughly 41,000 survey responses. It answers breakdowns by market, region, RSM, ASM, week, month and reason.
- **Why it is trustworthy.** The model cannot write SQL or run code. It fills a fixed, enumerated tool schema; pandas computes every figure; only aggregates are sent to the model, never raw customer records.
- **What has to change for company scale.** Three additions: a governed semantic layer so metrics are defined once, a document retrieval path with citations, and per-user permission filtering - which does not exist today.
- **Recommended shape.** One orchestrator over three retrieval paths (structured, documents, systems), with identity filtering, audit logging and an evaluation suite underneath all three.
- **Recommended sequencing.** Four phases over roughly four months, each independently useful. Scope by decision, not by data source.

## 1. What the current assistant does

The NPS assistant is neither a document search tool nor a natural-language-to-SQL system. It is a tool-calling agent over an in-memory table, and that narrower design is the reason its numbers can be relied on.

### How a question is answered

The dashboard loads the `silver.Hatch_NPS_Report` view from Azure SQL once every six hours, authenticating with an Azure AD service principal rather than a stored database password. The assistant receives the data already filtered to whatever the user has selected on screen, so its answers always match the dashboard in front of them. Each question then runs through three components:

- **A grounding summary** - overall NPS, promoter / passive / detractor split, the weeks in view and a per-country table, sent with every question.
- **A single tool** - the model cannot write a query. It fills a fixed schema: which dimensions to group by (chosen from a closed list of eight), optional filters, and a row limit. The application translates that into a pandas aggregation and returns a small table, capped at 40 rows.
- **A short agent loop** - up to five tool calls per question, with the last twelve conversation turns carried forward for context.

### The four properties worth preserving

| Property | How it is enforced today | Why it matters at company scale |
|---|---|---|
| **The model never does arithmetic** | Every figure is computed by pandas; the model only chooses what to compute. | This is what prevents invented numbers. It must survive the move to a warehouse. |
| **Raw records never leave the app** | Only aggregated results are sent to the model - no customer identifiers, no free-text comments. | Becomes a formal data-egress control once HR, finance and customer data are in scope. |
| **No injection surface** | The tool takes an enumerated list, not a query string. There is no path from a typed question to arbitrary SQL. | Generated SQL reintroduces this risk; it is contained with read-only credentials and limits. |
| **Answers match what is on screen** | The assistant is handed the filtered dataset, not the full table. | Generalises to a rule: retrieval is always scoped to the user's current context and rights. |

### Current limitations

- **One dataset, held in memory.** No joins, no second source. This does not extend past a few hundred thousand rows.
- **The dimension list is hard-coded.** The detractor follow-up feedback columns added in September are not yet exposed to the assistant, so it cannot answer "what do detractors say about DOC quality" even though the data is present. This is a days-long fix and the natural first step.
- **No per-user permissions.** Everyone who opens the dashboard sees everything. Acceptable for one NPS table; not acceptable for company-wide data.
- **No question log and no evaluation set.** There is currently no way to measure whether a change made answers better or worse.

## 2. Recommended architecture

Five layers. The orchestrator is the only component that talks to the user; it decides whether a question is numerical, documentary or both, and calls the matching path.

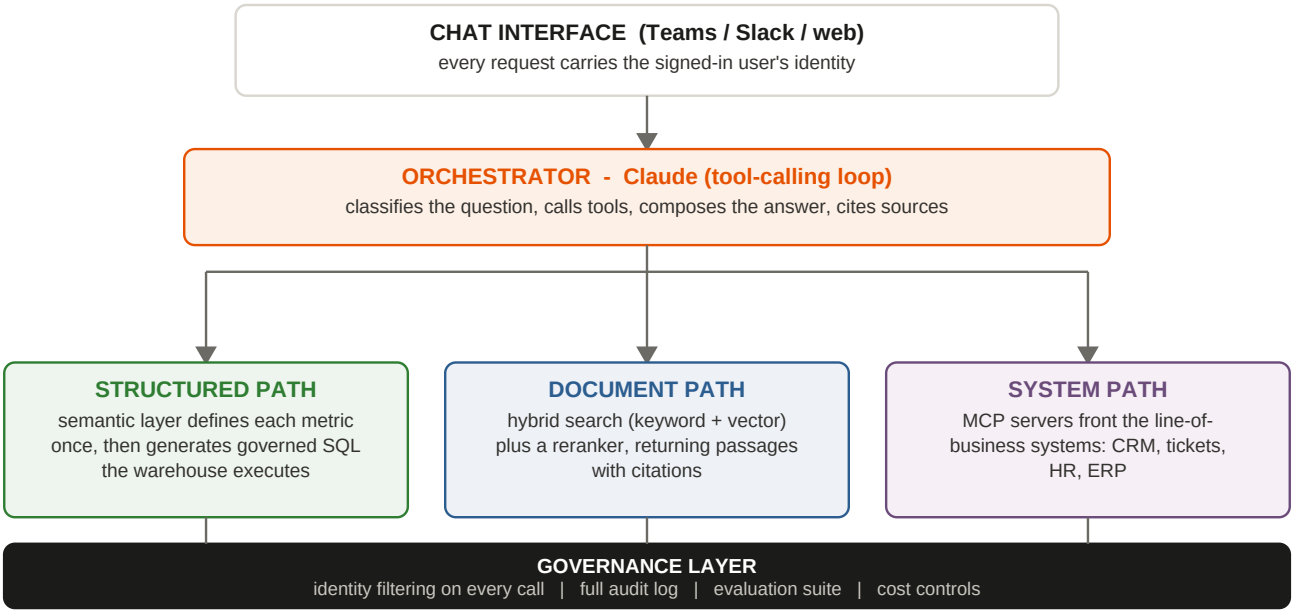


Figure 1 - One orchestrator over three retrieval paths. Governance is a layer, not a feature of any single path.

### Layer by layer

| Layer | Role | Recommended technology |
|---|---|---|
| **Orchestrator** | Plans the answer, calls tools, composes the result, attaches citations. | Claude Opus 5.5 for the main loop; Claude Haiku 4.5 for routing and classification. Prompt caching on the schema description, which is large and identical across requests. |
| **Structured path** | Answers numerical questions. A semantic layer defines each metric once, so NPS means the same thing everywhere. | Microsoft Fabric semantic model (shortest path given the existing Azure SQL estate); dbt Semantic Layer or Cube are equivalent alternatives. |
| **Document path** | Answers policy and knowledge questions from unstructured content, with sources attached. | Azure AI Search with hybrid retrieval - keyword and vector together - plus a reranker. Claude's citations feature returns the exact quoted text and its page location. |
| **System path** | Reaches line-of-business systems for live operational answers. | Model Context Protocol (MCP) servers - one integration standard rather than a bespoke connector per system. |
| **Governance** | Identity filtering, audit logging, evaluation, cost control. | Entra ID token propagation, row-level security in the warehouse, structured logging of every question, query and answer. |

### Why not simply let the model write SQL

Unconstrained natural-language-to-SQL is where most enterprise assistants quietly begin to mislead: the query runs, the number looks plausible, and nobody notices the join was wrong. A semantic layer sits between the model and the warehouse so that the model chooses a metric and dimensions - a structured choice, exactly like today's enumerated tool, only over hundreds of metrics instead of eight. Alongside it, a library of reviewed question-and-query pairs gives the model worked examples; this is the single highest-return investment in answer accuracy.

## 3. How accuracy is achieved, by question type

| Question type | Example | Mechanism | Guardrail |
|---|---|---|---|
| **Numerical** | What is NPS in Rwanda last month? | Semantic layer generates SQL; the warehouse computes the figure. | Show the query and the result table; one central metric definition. |
| **Documentary** | What is our parental leave policy? | Hybrid search and reranking return passages; the answer cites them. | Citations are mandatory; decline when retrieval is weak rather than paraphrase. |
| **Mixed** | Why did Rwanda's NPS drop, and what does policy say on compensation? | The orchestrator calls both paths and composes one answer. | Figures come from the warehouse, prose from the documents - never blended. |

> **Two things are not optional.** A library of reviewed question-to-query pairs, and an evaluation set of 100 to 200 real questions with known-correct answers, re-run on every change. Without the evaluation set there is no way to distinguish an improvement from a regression.

## 4. Security considerations

Today's assistant uses one service account and has no concept of the person asking. That design does not survive contact with company-wide data, and the changes below are prerequisites rather than enhancements.

> **Permissions must be enforced at retrieval, never in the prompt.** An instruction such as "do not show HR data to non-HR staff" is a suggestion to the model, not a control. The retriever must be incapable of returning what the user is not entitled to see.

- **Identity-aware retrieval.** Capture access rights when content is indexed and filter on the user's token at query time; apply row-level security in the warehouse.
- **Treat every retrieved document as untrusted input.** A file in SharePoint can contain instructions aimed at the model. Once the assistant can act as well as read, this becomes a genuine data-exfiltration route: keep any tool that writes or sends behind explicit human confirmation.
- **Least privilege on data access.** Read-only credentials, row limits, query timeouts and cost caps on everything the model can trigger.
- **Log every interaction** - question, tools called, query generated, rows returned, answer, user. This serves audit, debugging and doubles as the source of the evaluation set.
- **Settle data residency and retention with Legal before the pilot.** Zero data retention arrangements are available and are easier to agree up front than retrofit.
- **Move the API credential in-tenant.** The current key sits in Streamlit Cloud secrets, which is appropriate for one public dashboard but not for a company-wide assistant.

## 5. Delivery plan

Each phase is independently useful. Building all three retrieval paths at once is the common failure mode: it produces a demonstration that answers everything shallowly and nothing dependably.

| Phase | Timing | Scope | Proves |
|---|---|---|---|
| **1. Widen what works** | Weeks 1-2 | Expose the follow-up feedback columns; add two or three further curated tables; begin logging every question asked. | Immediate value at near-zero risk, and starts accumulating the evaluation set. |
| **2. Documents, one department** | Months 1-2 | Choose the department with the clearest permissions and the highest question volume. Hybrid search, citations, permission filtering. | That citations and access filtering work on a narrow, verifiable surface. |
| **3. Governed structured path** | Months 2-4 | Semantic layer over the warehouse, identity propagation, reviewed query library, evaluation suite in place. | That numerical answers hold up against known-correct results. |
| **4. Unify** | Month 4 onward | Single orchestrator across all paths, MCP for the long tail of systems, evaluations in continuous integration, cost dashboards. | That the assistant is operable and measurable as a product. |

### Indicative model costs

Published Anthropic API rates per million tokens, as of this brief. Model usage is typically the smaller line; search infrastructure, warehouse compute and engineering time dominate. Cached input - the schema and instructions resent with every question - is billed at roughly a tenth of the standard input rate, which is the largest single cost lever.

| Model | Role in the architecture | Input / 1M | Output / 1M |
|---|---|---|---|
| **Claude Opus 5.5** | Orchestrator and analytical reasoning | $4.00 | $20.00 |
| **Claude Sonnet 5.5** | High-volume routes where it holds quality | $2.00 | $10.00 |
| **Claude Haiku 4.5** | Routing, classification, cheap sub-tasks | $1.00 | $5.00 |

## 6. Recommendation

> **Scope the assistant by decision, not by data source.** The question "which fifteen questions do our leaders ask every week that take an analyst a day to answer?" will produce a far better system than "index everything". The NPS assistant is already a working instance of the correct pattern; the work ahead is governance and breadth, not a different architecture.

### Immediate next steps

- Approve Phase 1 and begin logging questions this month - the log becomes the evaluation set that every later phase depends on.
- Nominate the department for the Phase 2 document pilot, chosen on clarity of access rights rather than enthusiasm.
- Open the data residency and retention discussion with Legal now, in parallel with Phase 1.
- Name an owner for metric definitions. The semantic layer is a governance commitment before it is a technical one, and it fails without a custodian.

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Prepared from a review of the live dashboard codebase (data access layer, dashboard application and assistant module) and the current production dataset. Model identifiers, pricing and API capabilities verified against Anthropic platform documentation current at the date of this brief.